

IEDA 1250 Optimizing Decisions for Personal and Business Development

Exercises: Game Theory

Instructor: Bo YANG

Summer, 2026



For the game having the following payoff table, determine the optimal strategy for each player by successively eliminating dominated strategies. Indicate the order in which you eliminated strategies.

Strategy		Player 2		
		1	2	3
Player 1	1	-3	1	2
	2	1	2	1
	3	1	0	-2

Consider the game having the following payoff table.

Strategy		Player 2			
		1	2	3	4
Player 1	1	5	-7	-2	2
	2	-2	2	-5	5
	3	-2	5	-2	7

Determine the optimal strategy for each player by successively eliminating dominated strategies.

Find the saddle point for the game having the following payoff table for Player 1.

Strategy		Player 2		
		1	2	3
Player 1	1	1	2	6
	2	4	3	5
	3	5	1	0

Use the minimax criterion to find the best strategy for each player. Does this game have a saddle point? Is it a stable game?

Use the graphical solution to solve this zero-sum game.

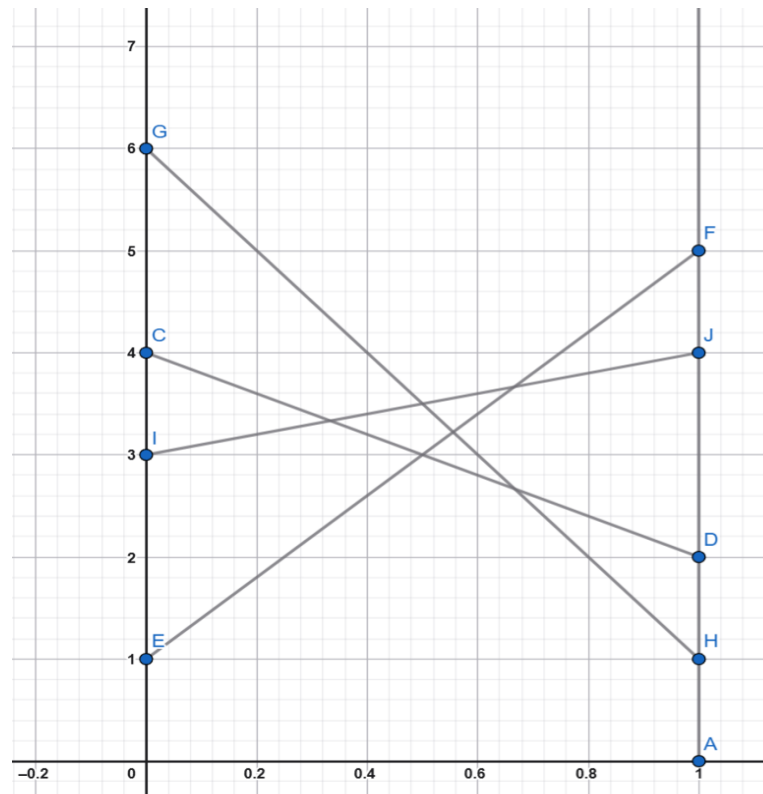
		Player B			
		B1	B2	B3	B4
Player A	A1	2	5	1	4
	A2	4	1	6	3

Step 1: We assign probability p and $1-p$ to strategies A1 and A2, respectively

Step 2: Suppose Player A uses p and $1-p$ for here two strategies, then she would expect:

- 2.1 If Player B uses B1, then her expected payoff is $2p + 4(1-p) = 4 - 2p$
- 2.2 If Player B uses B2, then her expected payoff is $5p + (1-p) = 1 + 4p$
- 2.3 If Player B uses B3, then her expected payoff is $p + 6(1-p) = 6 - 5p$
- 2.4 If Player B uses B4, then her expected payoff is $4p + 3(1-p) = 3 + p$

Step 3: Draw payoffs in a graph



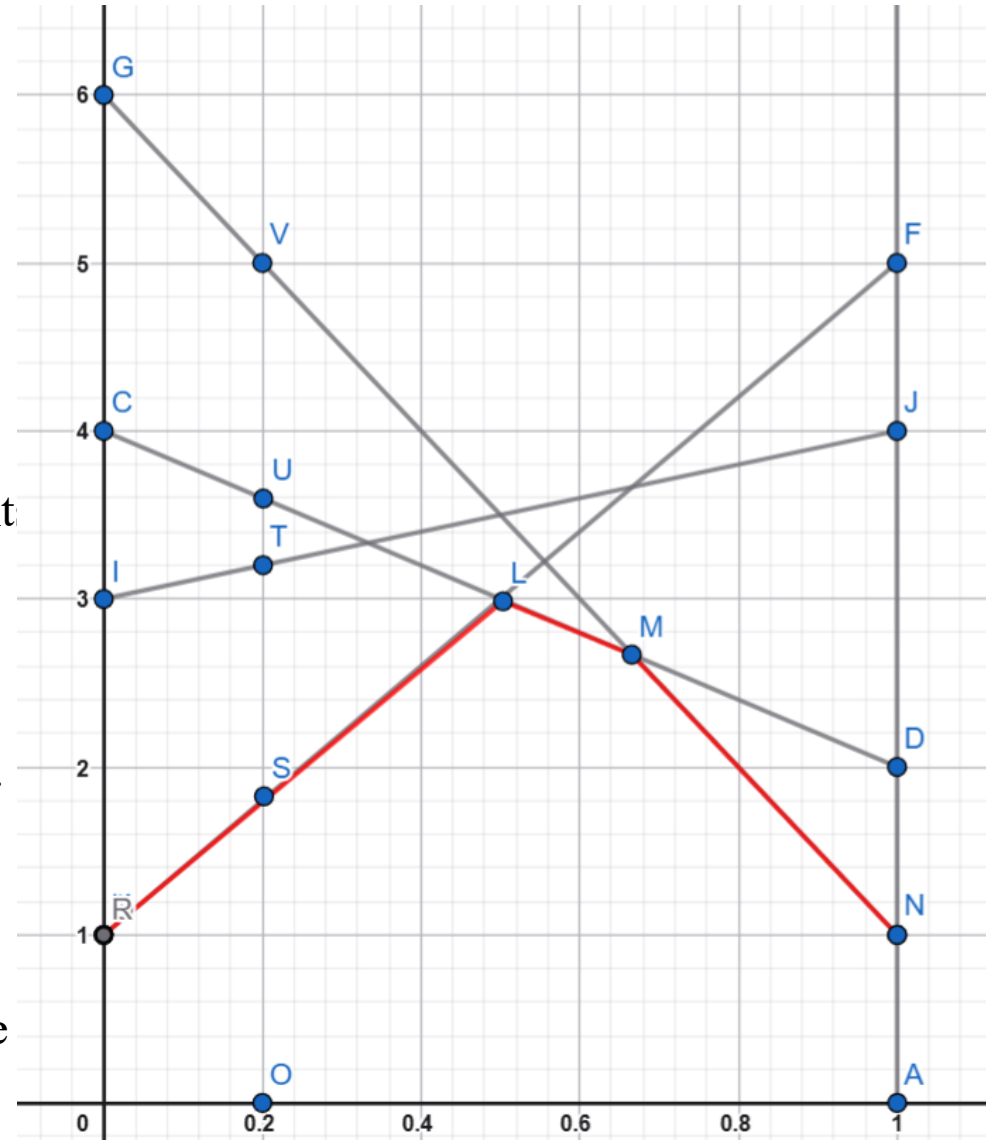
Step 4: Determine the optimal p value for Player A

Reasoning: Suppose Player A chooses $p=0.2$, i.e., point O in the figure. Then the four payoff functions indicate that the possible payoffs are 1.8 (point S), 3.2 (point T), 3.6 (point U), and 5 (point V).

However, since Player B plays against her, the actual payoff Player A can achieve is the smallest/minimum one among these four point. That is, the payoff 1.8 (point S)

Player A can keep varying the value of p . She will realize that the actual payoffs she can achieve are the red line segments. Any payoff above these red lines are impossible, because Player B plays against her.

Thus, Player A will choose the point L and the corresponding p value, because ...

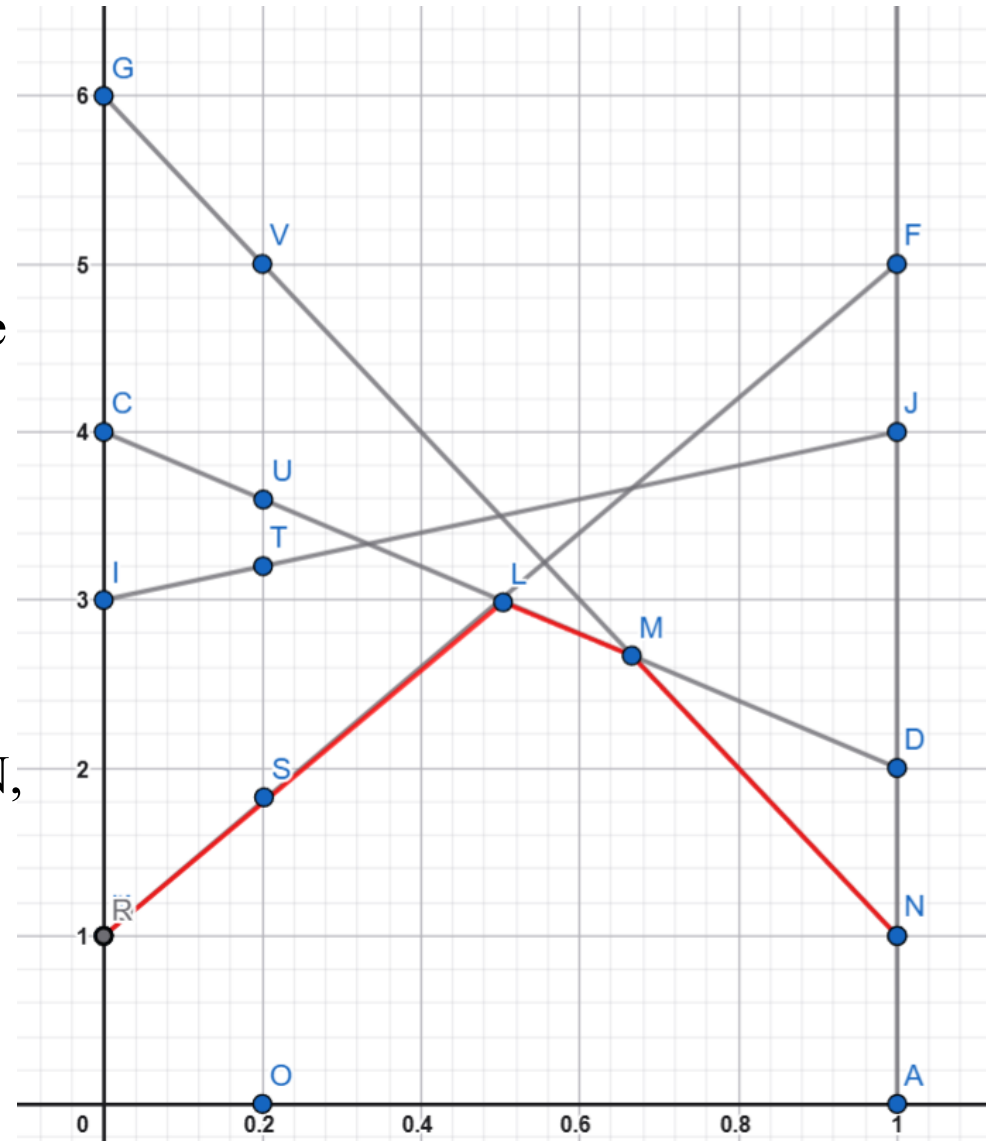


Reasoning (continued):

Thus, Player A will choose the point L and the corresponding p value, because it is the largest actual payoff she can obtain.

Observe that point L is an intersection of the line RF and CD. These two lines correspond to $1+4p$ and $4-2p$.

Step 5: Point L is not on the line IJ and GN, which correspond to $3+p$ and $6-5p$ and strategies B3 and B4. Therefore, $q_3=q_4=0$.



Player B

Player A

	B1	B2	B3	B4
A1	2	5	1	4
A2	4	1	6	3

Step 6: Now we consider Player B, i.e., using perspective of Player B. He assigns probability q to B1 and $1-q$ to B2.

Step 7: Then he would expect:

- If Player A chooses A1, his loss is $2q + 5(1-q)$
- If Player A chooses A2, his loss is $4q + (1-q)$

Step 8: Meanwhile, we know that Player A uses probability $1/2$ for each of her strategy. This suggests that these two strategies, A1 and A2, are equally good for her (or equally bad for Player B) at equilibrium.

Thus, the losses should be the same for Player B regardless of what strategies Player A uses, i.e.,
 $2q + 5(1-q) = 4q + (1-q)$

Use the graphical solution to solve this zero-sum game.

		Player B	
		B1	B2
Player A	A1	1	5
	A2	4	0
	A3	0	7
	A4	2	4

		Player B	
		B1	B2
Player A	A1	1	5
	A2	4	0
	A3	0	7
	A4	2	4

Step 2: Suppose Player B uses q and $1-q$ for the two strategies, then he would expect:

- If Player A uses A1, then his expected loss is $q + 5(1-q) = 5 - 4q$
- If Player A uses A2, then his expected loss is $4q$
- If Player A uses A3, then his expected loss is $7 - 7q$
- If Player A uses A4, then his expected loss is $2q + 4(1-q) = 4 - 2q$

Suppose Player B uses q and $1-q$ for the two strategies, then he would expect:

- If Player A uses A1, then his expected loss is
 $q + 5(1-q) = 5 - 4q$
- If Player A uses A2, then his expected loss is
 $4q$
- If Player A uses A3, then his expected loss is
 $7 - 7q$
- If Player A uses A4, then his expected loss is
 $2q + 4(1-q) = 4 - 2q$

